Hierarchical Tests of Serial Independence Using Coherent Fock States

Probabilistic construction, permutation calibration, simulations, and comparative audit

Technical research report

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10 November 2026 | Version 1.0 | Python code and data included

Editorial status: exploratory preprint draft, not peer reviewed. The mathematical connections to existing kernel and independence tests preclude claiming demonstrated novelty for this procedure. Conclusions are based on simulation, not on a theorem establishing superior power.

Abstract

We study a serial-independence test inspired by the hierarchy of coherent Fock states. Each observation is rank-transformed and represented by a truncated vector of occupation-basis amplitudes; interactions across two, three, and four time instants are measured using tensor norms and maximum coefficients. The null distribution is estimated by permuting the series, which provides finite-sample validity under exchangeability with a prespecified configuration. A new evaluation of 990 primary simulated series across eleven scenarios, supplemented by 300 fourth-order series, 275 sparse-alternative series, 300 bandwidth-sensitivity series, and 200 size-audit series, detects ARCH, bilinear dependence, and a purely ternary dependence missed by Ljung–Box. However, BDS matches or exceeds its power in the cases tested. An exploratory maximum-modal-statistic variant recovers power against concentrated fourth-order dependence, but a simple sign-product control performs similarly. The method is interpretable and reproducible; sufficient statistical originality and broad superiority over BDS, HSIC/dHSIC, or AutoHSIC have not been demonstrated. We provide derivations, limitations, Python code, and an empirical illustration.

Keywords: Fock space; coherent states; strong white noise; serial independence; nonlinear dependence; tensors; permutation tests; BDS; HSIC.

Repository: The accompanying ZIP archive contains the LaTeX source, scripts, predictions, random seeds, and CSV files. No external hosting is assumed.

1 Scientific question and scope

The original proposal was to use a hierarchy of Fock modes to identify serial dependence invisible to linear correlations. The precise statistical question is whether a sequence $(X_t)_{t=1}^N$ is independent and identically distributed:

$$H_0: \quad P(X_1 \in A_1, \ldots, X_N \in A_N) = \prod_{t=1}^{N} P(X_t \in A_t) \quad \text{for all } A_t. \tag{1}$$

Weak white noise only requires constant mean, constant finite variance, and zero covariance at every nonzero lag. ARCH provides a standard counterexample: a process may be uncorrelated while

its conditional variance depends on the past [2]. Our test targets (1), **not** the mere absence of autocorrelation, nor a certification of cryptographic randomness or deterministic chaos.

Contributions and limitations. We provide a concrete modal description of independence across several temporal orders, a reproducible permutation algorithm, a power study with linear and nonlinear controls, a sparse-interaction variant, and a scalability audit. We do not establish that the test is statistically new: its Fock feature construction converges to the Gaussian kernel. Related work includes BDS [3], bicorrelation tests [4], generalized spectra [5], kernel independence methods [6, 7, 8], and AutoHSIC [9].

2 Foundations: from a Fock basis to a feature kernel

2.1 Real coherent states and occupation numbers

Let $\mathcal{F} = \operatorname{span}\{|n\rangle\}_{n\geq 0}$ be a bosonic Fock space with an orthonormal basis. The operators $a, a^\dagger$ satisfy $[a, a^\dagger] = 1$. A real coherent state with parameter α is

$$|\alpha\rangle = e^{-\alpha^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \qquad a|\alpha\rangle = \alpha|\alpha\rangle, \qquad \langle\alpha|\alpha\rangle = 1. \tag{2}$$

For $z \in \mathbb{R}$ and bandwidth $\sigma > 0$, set $\alpha = z/\sigma$. Truncation at occupation order K gives the feature map

$$\psi_K(z) = \left(e^{-\alpha^2/2}, e^{-\alpha^2/2}\alpha, \ldots, e^{-\alpha^2/2} \frac{\alpha^K}{\sqrt{K!}} \right) \in \mathbb{R}^{K+1}. \tag{3}$$

The inner product of two finite feature maps is

$$\langle \psi_K(z), \psi_K(w) \rangle = e^{-(z^2+w^2)/(2\sigma^2)} \sum_{n=0}^{K} \frac{(zw/\sigma^2)^n}{n!}. \tag{4}$$

Since the exponential series converges absolutely, increasing K yields exactly

$$\lim_{K\to\infty} \langle \psi_K(z), \psi_K(w) \rangle = \exp[-(z-w)^2/(2\sigma^2)], \tag{5}$$

that is, the Gaussian kernel. **This is not a fundamentally new independence construction**: it is a finite, interpretable approximation indexed by occupation levels. The norm of the truncated state depends on z; we do not renormalize each feature vector separately, because doing so would alter the kernel in (4).

2.2 Rank transformation and centering

We use $u_t = (\operatorname{rank}(X_t) - 1/2)/N$ and $z_t = \Phi^{-1}(u_t)$, using average ranks for ties. This limits the influence of marginal tails and is equivariant under permutations. The features in (3) are centered and, in the main implementation, each coordinate with nonzero standard deviation is divided by its sample standard deviation:

$$f_{t,n} = \frac{\psi_{K,n}(z_t) - \bar{\psi}_n}{s_n}, \qquad f_{t,n} = 0 \text{ if } s_n \approx 0. \tag{6}$$

This standardization makes modes more comparable but changes the effective kernel in (5): exact Gaussian-kernel equivalence holds for the *unweighted* version. The map still defines a positive semidefinite kernel, and the permutation test retains its level under H_0.

3 Dependence statistics across two, three, and four times

3.1 Two-time modal covariances

With $d = K + 1$ and a prespecified maximum of five lags, define

$$\widehat{C}_{nm}(\ell) = \frac{1}{N-\ell} \sum_{t=\ell+1}^{N} f_{t,n} f_{t-\ell,m}, \tag{7}$$

$$T_2 = \sum_{\ell=1}^{5} \frac{N-\ell}{d^2} \sum_{n,m=0}^{K} \widehat{C}_{nm}(\ell)^2. \tag{8}$$

Linear dependence may be visible in low-order coordinates; heteroskedasticity or nonlinear pairwise dependence may appear as coupling between higher-order coordinates.

3.2 Joint temporal interactions

The third-order tensor uses consecutive times:

$$\widehat{M}^{(3)}_{nmp} = \frac{1}{N-2} \sum_{t=3}^{N} f_{t,n} f_{t-1,m} f_{t-2,p}, \qquad T_3 = \frac{N-2}{d^3} \|\widehat{M}^{(3)}\|_F^2. \tag{9}$$

The four-time extension is

$$\widehat{M}^{(4)}_{nmpq} = \frac{1}{N-3} \sum_{t=4}^{N} f_{t,n} f_{t-1,m} f_{t-2,p} f_{t-3,q}, \qquad T_4 = \frac{N-3}{d^4} \|\widehat{M}^{(4)}\|_F^2. \tag{10}$$

These quantities are not occupation moments of a financial-market generator: they are temporal interactions among Fock-inspired features. Their purpose is diagnostic. By construction, the cost scales roughly as $O(BNK^q)$ for an order-q tensor with B permutations, and tensor storage scales as $O(K^q)$. Temporal lags and orders are finite; **none of the tests with $q \le 4$ detects every alternative to H_0.**

3.3 Variant for sparse interactions

When a single tensor component carries almost all the signal, the Frobenius norm distributes its influence across d^q terms. We therefore consider, as an **exploratory** variant,

$$T_q^{\max} = (N-q+1) \max_{0 \le n_1,\dots,n_q \le K} \left| \widehat{M}^{(q)}_{n_1 \cdots n_q} \right|^2, \qquad q \in \{3,4\}. \tag{11}$$

Unlike T_q, this statistic favors sparse alternatives; calibration must account for maximization over all components. A simple sign-product test, also calibrated by permutation, provides an appropriate targeted control.

4 Calibration, guarantees, and multiplicity

Proposition 1 (Finite-sample permutation validity)**.** *Fix K, σ, q, the lag set, and the statistic T, and let T_{obs} be its value on $X_1, \dots, X_N$. If the i.i.d. null hypothesis holds and B independent uniformly random index permutations are drawn, then*

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T_b \ge T_{\text{obs}}\}}{B+1} \tag{12}$$

is a valid Monte Carlo p-value (superuniform, conservative in the presence of ties) for testing H_0. No Gaussian assumption or existence of higher moments of X_t is required.

Proof. Under H_0, the joint data distribution is invariant under every time permutation. Conditional on the observed multiset, the original ordering is exchangeable with the randomly permuted orderings. Thus, the rank of the original statistic among the $B + 1$ values is uniform, or conservative when ties occur. Centering, ranking, and standardization depend only on the multiset and preserve this invariance. Expression (12) is the adjusted upper-tail rank proportion. □

Multiple components. The primary test, specified before evaluating the main scenarios, combines T_2 and T_3 with $K = 4, \sigma = 1.4$ using $p_{\text{prim}} = \min\{1, 2\min(p_2, p_3)\}$. This Bonferroni correction controls familywise error; results for each component are also shown separately and exploratorily. Searching for the best K, q, or σ *after* examining the same series requires correction over the full search. With 99 permutations the smallest resolvable p-value is 0.01: Bonferroni correction across six tests cannot reach 0.05 (the minimum adjusted value is 0.06). Accordingly, **we do not report rejection rates for the exploratory six-test omnibus** at this resolution; at least 199 permutations or joint calibration of the search is required.

Warning about residuals. Simple permutation is not automatically valid when features are computed from residuals of a model whose parameters were estimated on the same data. The null distribution must then account for fitting, for example by repeating estimation within each resample. Nor is validity guaranteed under time heterogeneity, trends, or regime shifts: such alternatives violate the i.i.d. null, and rejection is not a false positive.

5 Analytical higher-order alternatives

Let A, B, C be independent Rademacher variables (taking ± 1 with equal probability). The block (A, B, AB) has *pairwise independent* components but satisfies $A \cdot B \cdot (AB) = 1$. In the block (A, B, C, ABC) all subsets of at most three components are independent, while the product of all four is one. We concatenate independent blocks, randomize the initial phase uniformly to obtain stationarity, and add independent Gaussian noise with standard deviation 0.15 so that tests for continuous data apply. The noise preserves independence among proper subsets and makes the problem less trivial.

This construction explains why T_2 alone cannot detect XOR-3 and why neither T_2 nor T_3 can systematically detect XOR-4. The limitation concerns the *order of temporal interaction*, not only the number of modes K. Actual power also depends on whether a chosen statistic concentrates or dilutes the signal.

6 Experimental design and competitors

6.1 Simulated data and sample sizes

The new main battery consists of **990 series**: eleven mechanisms, 55 independent replications at $N = 600$ and 35 at $N = 1500$. We use 99 permutations per series and $K \in \{2, 4, 8\}$, with $\sigma = 1.4$. The fourth-order battery adds 300 series with $K \in \{2, 4\}$; the sparse variant adds 275 independent series with $K = 4$ and 199 permutations; the bandwidth-sensitivity battery adds 300 series; and an independent size audit of the fourth-order maximum modal statistic adds 200 i.i.d.-null series. The two maximum-modal experiments use 199 permutations. Sign-based controls are applied to the same 990 series as the main battery and are **not** counted as new replications. There are 2065 newly simulated series in the five programs, with different evaluation intensities. Individual files record random-generation seeds or deterministic reproducibility rules.

The processes include Gaussian i.i.d., Student-t_3 i.i.d., centered exponential i.i.d., Bernoulli i.i.d., AR(1) ($\rho = 0.3$), ARCH ($h_t = 0.35 + 0.65X_{t-1}^2$), bilinear $X_t = \varepsilon_t \varepsilon_{t-1}$, noisy XOR-3 and XOR-4,

persistent stochastic volatility, and a deterministic variance break. The latter is *not* a valid null case: it probes sensitivity to nonstationarity.

6.2 Benchmark tests

We compute Ljung–Box using 10 lags [1], ARCH–LM using five lags [2], and BDS with embedding dimensions 2–4 and Bonferroni correction across dimensions [3]. BDS is omitted for the strictly Bernoulli series because of ties and the conditions underlying its asymptotic approximation; XOR-3 and XOR-4 include continuous noise. For higher temporal order we also include a targeted permutation test of sign products. Rejection frequencies are Monte Carlo estimates, each with binomial uncertainty displayed using 95% Wilson intervals.

7 Statistical power, size, and number of modes

7.1 Primary comparison

Table 1: Empirical power/size at the 5% level, $N = 600$; 55 replications per row. Individual Fock tests without combination; the last column uses the primary correction with $K = 4$, $q = 2, 3$.

Scenario	Ljung–Box	ARCH–LM	BDS(4)	Fock T_2	Fock T_3	Primary Fock
Gaussian i.i.d.	4%	2%	5%	7%	0%	0%
Student-t i.i.d.	2%	5%	0%	4%	4%	2%
Exponential i.i.d.	5%	9%	2%	4%	4%	2%
Bernoulli i.i.d.	7%	5%	—	7%	2%	4%
AR(1)	100%	33%	85%	96%	44%	95%
ARCH	36%	100%	100%	100%	100%	100%
Bilinear	11%	100%	100%	100%	82%	100%
Noisy XOR-3	4%	2%	100%	5%	42%	20%
Noisy XOR-4	11%	5%	100%	7%	9%	5%
Latent volatility	22%	95%	95%	93%	38%	96%
Scale break	22%	100%	100%	100%	29%	100%

Table 2: The same rejection rates for $N = 1500$; 35 replications per row.

Scenario	Ljung–Box	ARCH–LM	BDS(4)	Fock T_2	Fock T_3	Primary Fock
Gaussian i.i.d.	14%	3%	3%	6%	6%	6%
Student-t i.i.d.	0%	3%	6%	3%	9%	6%
Exponential i.i.d.	6%	11%	9%	6%	9%	3%
Bernoulli i.i.d.	3%	0%	—	0%	0%	0%
AR(1)	100%	74%	100%	100%	89%	100%
ARCH	49%	100%	100%	100%	100%	100%
Bilinear	11%	100%	100%	100%	100%	100%
Noisy XOR-3	3%	0%	100%	3%	100%	94%
Noisy XOR-4	3%	3%	100%	0%	6%	0%
Latent volatility	14%	100%	100%	100%	54%	100%
Scale break	29%	100%	100%	100%	43%	100%

Size control. Under four i.i.d. nulls (Gaussian, Student, asymmetric, and discrete), the primary test shows rejection rates compatible with the nominal 5%, given the limited number of replications.

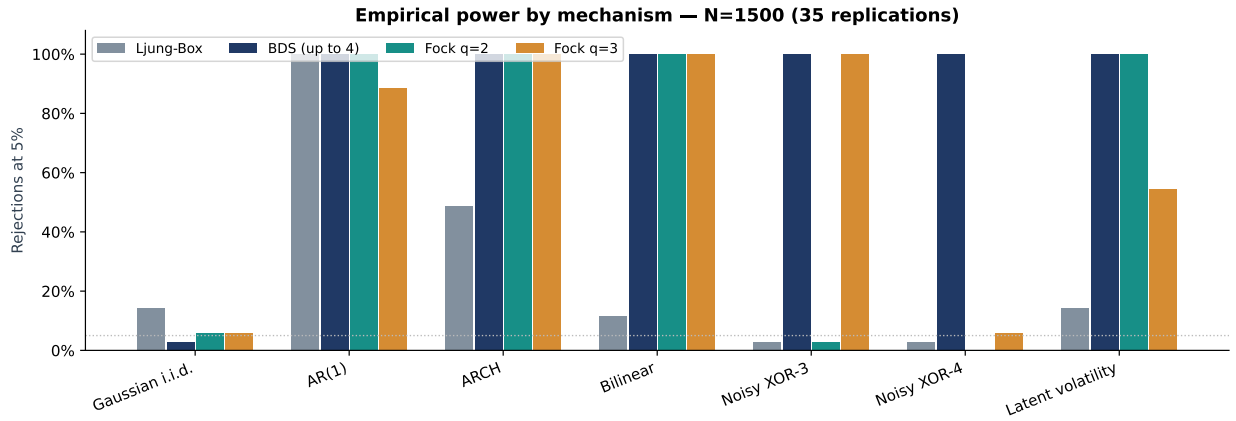


Figure 1: Comparison with established tests: the hierarchy detects some nonlinear dependencies missed by Ljung–Box, but does not generally outperform BDS.

For $N = 1500$, 35 Gaussian simulations produced a 14% rejection rate for Ljung–Box, a fluctuation that does not establish a systematic size distortion at such low precision. The permutation test makes no normality assumption. Individual percentages must be interpreted with binomial confidence intervals, not as if they were exact population rates.

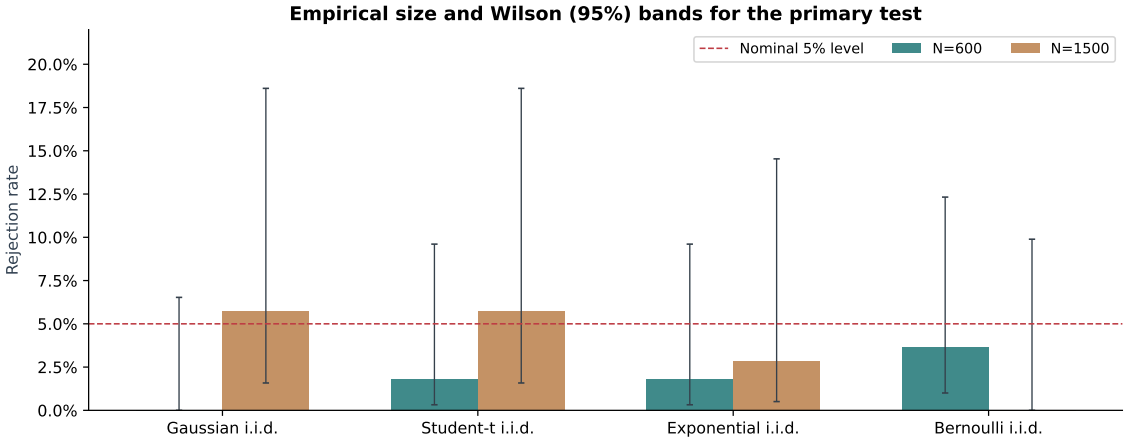


Figure 2: Simulated size of the primary test with 95% Wilson bands. These summarize Monte Carlo replications, not coverage intervals for one observed series.

7.2 Mode order and coherent-state bandwidth

7.3 Fourth-order dependence and sparse alternatives

The initial four-time Frobenius extension performed poorly on XOR-4: at $N = 1500$ it rejected approximately 10% of the series in a dedicated battery of 20 replications. In the subsequent extension, designed *after inspecting those results*, replacing the norm by the maximum component raised XOR-4 power to 100% in 20 replications, and likewise reached 100% for XOR-3 in 20. This suggests a potential *family of diagnostics for sparse interactions*, but a simple sign-product control also achieved 100% for both mechanisms and sample sizes; BDS likewise reached 100% for XOR-3/XOR-4 in the primary experiment.

7.4 Independent size audit of the maximum statistic

The exploratory fourth-order maximum study initially showed an unusually high rejection frequency for one i.i.d. cell with only 35 replications. To check the calibration, we ran another **200 independent**

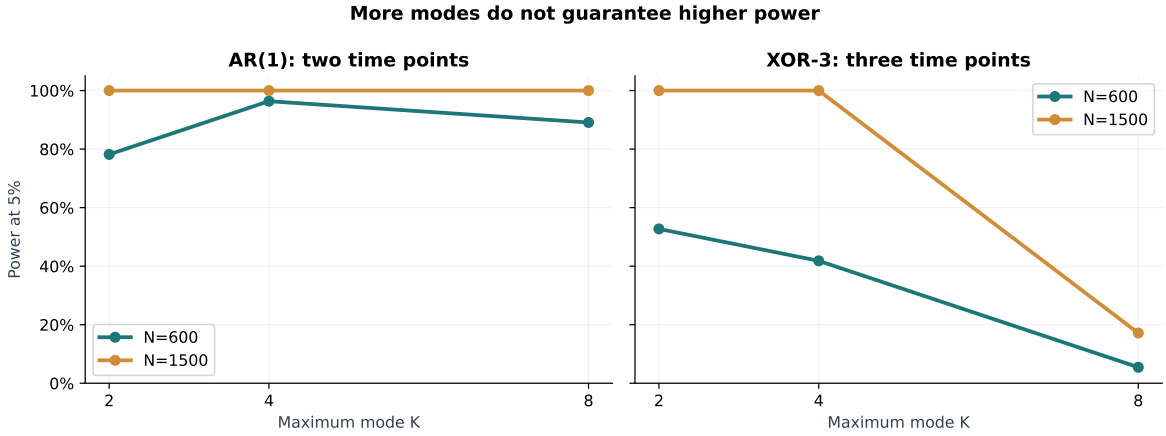


Figure 3: Power does not increase uniformly as K grows from 2 to 4 to 8, even at fixed sample size.

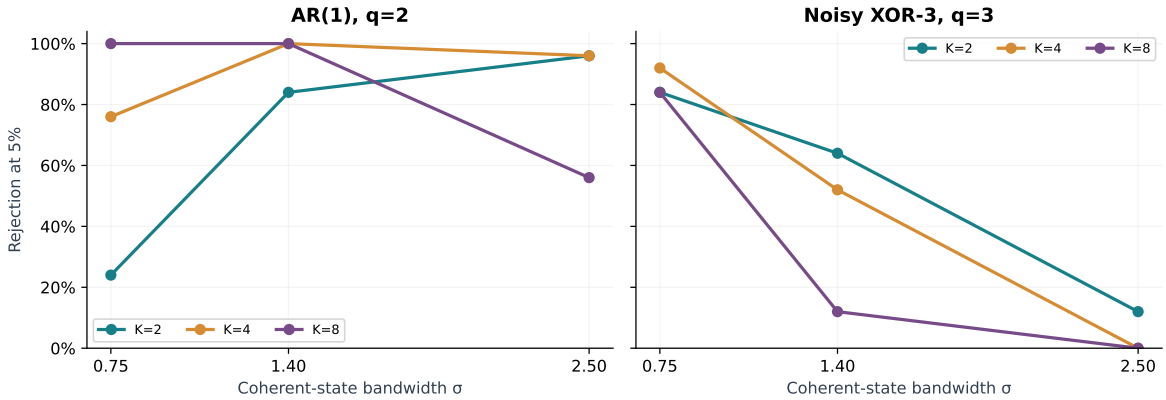


Figure 4: Independent sensitivity battery: $N = 600$, 25 replications per setting. Bandwidth is a genuine tuning parameter and must not be optimized on the test series without recalibration.

Table 3: Exploratory study of the modal maximum against the L^2 norm and the sign-product control. Power refers to the corresponding interaction order, with no post-hoc adaptive choice of q.

Scenario	N	L^2 modal	Modal maximum	Targeted signs	BDS(4)
Noisy XOR-3	600	26%	100%	100%	100%
Noisy XOR-4	600	0%	43%	100%	100%
Noisy XOR-3	1500	95%	100%	100%	100%
Noisy XOR-4	1500	15%	100%	100%	100%

Note: BDS results come from the main experiment, which used other replications and sample choices; these columns do not constitute a paired test of superiority.

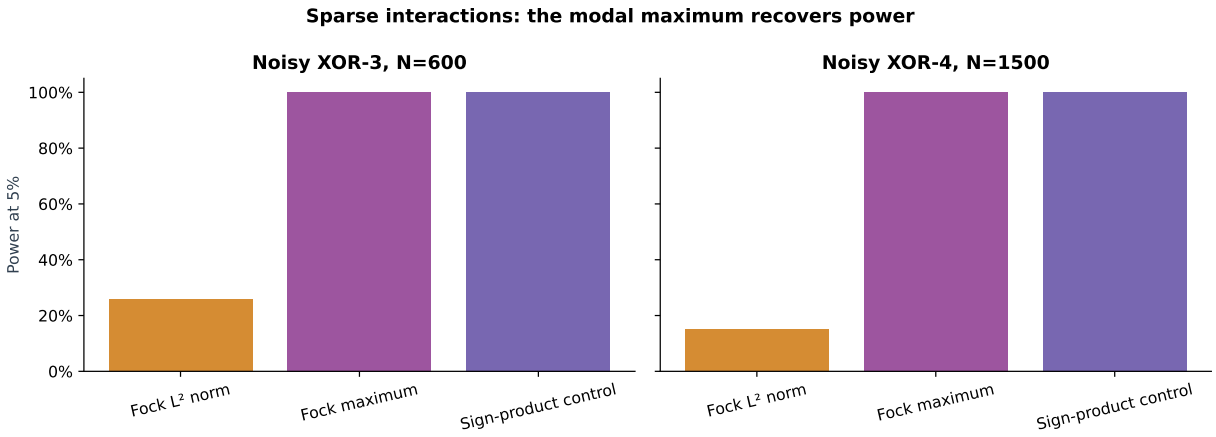


Figure 5: The modal maximum is more sensitive to very sparse dependencies; the much simpler sign control matches its performance for the selected alternatives.

series, 100 Gaussian and 100 Student-t_3, of length $N = 600$, with $K = 4$ and $B = 199$ permutations. The maximum-modal statistic $T_4^{\max}$ rejected **3/100 (3%)** under Gaussian noise and **7/100 (7%)** under Student noise. The corresponding 95% Wilson intervals were $[1.0\%, 8.5\%]$ and $[3.4\%, 13.7\%]$, both including the nominal 5% level. The added control is consistent with the permutation guarantee and suggests that the initially high rate reflected Monte Carlo variability. It does not replace validation using hundreds or thousands of replications when high-precision size estimates are required.

8 Illustrative application: daily returns and GJR residuals

We inspected 2481 daily Microsoft returns (2008–2017) retained from an earlier numerical investigation. The data were reused from a secondary historical dataset and are **not** an independently downloaded or externally validated financial series. On raw returns, the primary Fock test gives $p = 0.004$; on the series rescaled by estimated GJR–GARCH variance, it also gives $p = 0.004$. On rescaled residuals, Ljung–Box gives $p \simeq 0.285$, ARCH–LM gives $p \simeq 0.980$, and BDS corrected up to embedding dimension 4 gives $p \simeq 0.006$. This difference illustrates the potential utility of nonlinear diagnostics, *not* Fock superiority: BDS also rejects.

Permutation inference on estimated residuals must be regarded as **descriptive**: estimating GJR parameters and possible temporal breaks prevent direct application of the exact theorem in Section 4. A publishable residual analysis would need simulation or resampling under the fitted null model and refitting it within every replication.

9 Computational cost, methodological issues, and publication requirements

In these measurements the modal method is slower than BDS, but **the comparison does not match calibration procedures**: Fock uses 199 permutations, whereas BDS uses an asymptotic approximation. The table records the costs of these implementations, not a proof of Fock inefficiency in every configuration. Costs increase sharply with tensor order. The hierarchy may be interpretable, but efficiency has not been demonstrated. The maximum statistic may be more competitive when the signal is sparse, though it requires parameter selection and familywise calibration without optimization bias.

Limitations preventing a claim of superiority.

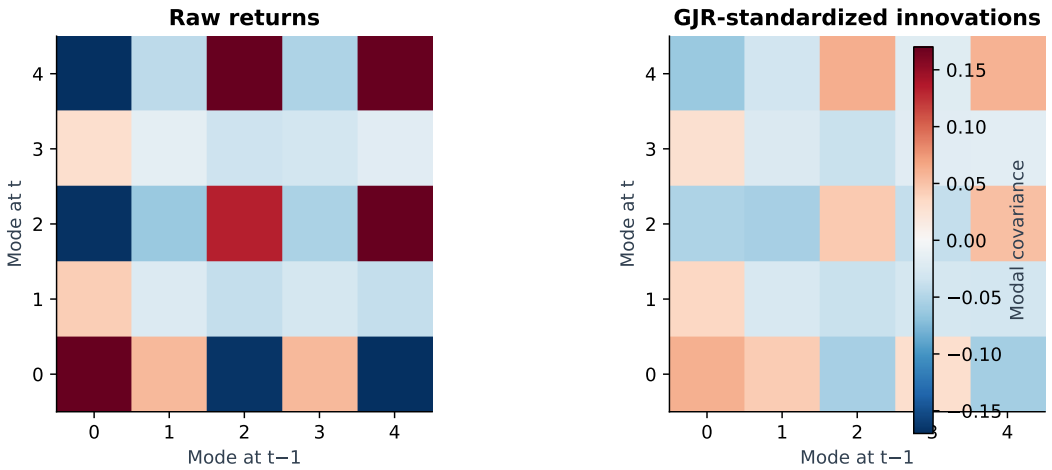


Figure 6: Lag-one modal covariance matrix. Its structure changes considerably after volatility rescaling, but rejection requires proper calibration when the scale has been estimated.

Table 4: Measured CPU times, median of three executions with one numerical thread. Time per series; Fock includes 199 permutations.

N	Method	Time (s)
600	Ljung–Box y ARCH–LM	0.001
600	BDS up to dim. 4	0.014
600	Fock q2/q3; $K = 2, 4, 8$; 199 perm.	0.261
600	Fock q2/q3/q4; $K = 2, 4$; 199 perm.	0.719
1500	Ljung–Box y ARCH–LM	0.001
1500	BDS up to dim. 4	0.081
1500	Fock q2/q3; $K = 2, 4, 8$; 199 perm.	0.588
1500	Fock q2/q3/q4; $K = 2, 4$; 199 perm.	1.612
2500	Ljung–Box y ARCH–LM	0.002
2500	BDS up to dim. 4	0.286
2500	Fock q2/q3; $K = 2, 4, 8$; 199 perm.	0.974
2500	Fock q2/q3/q4; $K = 2, 4$; 199 perm.	2.613

1. Gaussian kernels and HSIC/dHSIC/AutoHSIC tests already exist. A finite coherent-state representation alone does not substantiate novelty.
2. Choices of K, σ, q and lags materially affect power. No test using finitely many modes and finite temporal order is universally omnibus against all alternatives.
3. The main simulations contain only 35–55 replications per cell; differences of a few percentage points are inconclusive. The sparse extension was explicitly designed after inspecting previous results.
4. The experiment does not isolate *weak white noise*. It can reject marginal changes and heteroskedasticity even when linear autocorrelations vanish.
5. Permuting fitted residuals does not automatically account for parameter uncertainty in an estimated model or make the test robust to nonstationarity.
6. The $q = 4$ extension and searches over many modes may be computationally costly. The minimum attainable p-value depends on B and multiplicity correction.
7. There is no complete comparison with optimized implementations of AutoHSIC and recent multi-time kernel tests; BDS and the sign-product control already constrain originality claims.

10 Conclusions and publication assessment

1. **The procedure works as a diagnostic of i.i.d. serial independence.** Permutation calibration has rigorous justification under exchangeability and avoids assuming a Gaussian marginal distribution.
2. **Temporal interaction order matters.** Three-variable dependence can occur when every pair is independent; four-variable dependence may evade lower-order tests.
3. **The maximum modal statistic is promising for sparse alternatives.** We observe a large recovery of power against noisy XOR-3 and XOR-4, but this variant remains exploratory.
4. **There is no evidence of a general advantage over existing tests.** BDS matched or exceeded primary results, and simple targeted controls matched the best high-order results. No speed advantage was demonstrated.

Editorial assessment. This work can appropriately be deposited as an *exploratory technical note or reproducible preprint*, with proper attribution to prior work. It is **not** recommended for submission as a paper claiming a new, superior test without additional asymptotic theory, independent confirmatory replication, and equally calibrated comparisons with AutoHSIC/dHSIC.

11 Reproducibility and package structure

The package contains code that generates all synthetic series from recorded seeds and produces per-replication CSV files, tables, and figures. No external datasets are used in the simulations. The empirical Microsoft dataset is included in `msft_oos_returns.csv`, with the provenance warning stated above. Reproduction commands follow:

```
python -m pip install -r requirements.txt
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 MKL_NUM_THREADS=1
python fock_white_noise_study.py --suite main --sizes 600 --reps 55 --perms 99 --with-bds --out
    main600
python fock_white_noise_study.py --suite main --sizes 1500 --reps 35 --perms 99 --with-bds --out
    main1500
python fock_white_noise_study.py --suite quad --sizes 600 --reps 30 --perms 99 --with-bds --out
    quad600
```

```
python fock_white_noise_study.py --suite quad --sizes 1500 --reps 20 --perms 99 --with-bds --out
    quad1500
python sparse_modal_followup.py --n 600 --reps 35 --perms 199 --out sparse600
python sparse_modal_followup.py --n 1500 --reps 20 --perms 199 --out sparse1500
python higher_order_controls.py --n 600 --reps 55 --perms 199 --out sign_control
python higher_order_controls.py --n 1500 --reps 35 --perms 199 --out sign_control
python fock_white_noise_study.py --suite bandwidth --sizes 600 --reps 25 --perms 99 --out bandwidth
python empirical_demo.py
python timing_bench.py
python visuals_and_results.py
```

Computing note. The scripts reproduce the tables in Python environments using the versions in `requirements.txt`. Execution times depend on the CPU, BLAS, and operating system. The sensitivity experiment is exploratory, and timing values are not universal.

A Supplementary table: dependence by feature order

Scenario	N	$T_2(K2)$	$T_2(K4)$	$T_2(K8)$	$T_3(K2)$	$T_3(K4)$	$T_3(K8)$
AR(1)	600	78%	96%	89%	18%	44%	42%
AR(1)	1500	100%	100%	100%	43%	89%	66%
Noisy XOR-3	600	5%	5%	11%	53%	42%	5%
Noisy XOR-3	1500	0%	3%	0%	100%	100%	17%
ARCH	600	100%	100%	100%	100%	100%	100%
ARCH	1500	100%	100%	100%	100%	100%	100%
Gaussian i.i.d.	600	5%	7%	4%	4%	0%	7%
Gaussian i.i.d.	1500	6%	6%	9%	9%	6%	0%

B Record of the earlier pilot study

Before this battery, a separate pilot had been conducted with $N = 500$ (90 replications per scenario) and $N = 1500$ (45 replications per scenario), 99 permutations, and $K = 2, 4, 8$. In that pilot, the XOR-3 process was *discrete and had no added noise*; the third-order statistic also aggregated three lag combinations, rather than only the consecutive triplet used in the primary study. With $K = 4$, the third-order test rejected 100% of XOR-3 series at both sample sizes; Ljung–Box stayed near nominal size and BDS also detected it in all samples of that separate run. This looks more favorable than the noisy study, but **the results cannot be directly compared**: the observation mechanism and precise definition of T_3 changed. The archive includes `legacy_pilot/white_noise_fock.py` and prior CSV files so that the development history can be audited without hiding changes of specification. This pilot was not counted among the 2065 newly simulated series.

C Additional proof for XOR constructions and their limitations

In the three-variable block, A, B, and AB are Rademacher variables. For example, $P(A = 1, AB = 1) = P(A = 1, B = 1) = 1/4 = P(A = 1)P(AB = 1)$. The same holds for all other pairs. Nevertheless, $\mathbb{E}[A \cdot B \cdot AB] = \mathbb{E}[A^2B^2] = 1$. The mixture of phases of concatenated blocks assigns an independent uniform phase. Over three consecutive times, the triple-product identity appears in one-third of the phase alignments; the remaining alignments generally do not have the same moment. In the four-variable block (A, B, C, ABC), every subset of three observables is independent because one independent sign variable always remains, but $\mathbb{E}[A \cdot B \cdot C \cdot ABC] = 1$. Independent additive Gaussian noise preserves independence of lower-order subsets while attenuating the product signal. These constructions illustrate the difference between *increasing* K (the feature resolution of one observation) and *increasing* q (the order of temporal interaction).

D What would be needed for an original methodological article

An original advance beyond the kernel–Fock identity would require at least one of the following: (i) an adaptive rule for selecting modes and orders with exact or asymptotic familywise control and lower cost; (ii) a characterization of alternatives on which the hierarchical closure provably gains statistical power; (iii) an algorithm for evaluating high-order tensors without K^q growth; or (iv) replicated benefits over AutoHSIC, dHSIC, BDS, and higher-order tests on real-variable problems. This report establishes none of these four advances.

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